

3D printed rocket parts

NASA recently tested the most complex rocket engine parts ever designed by them, which were designed with additive manufacturing, or 3D printing, according to an IANS report. Apart from helping engineers build and test a rocket injector with a unique design, additive manufacturing also enabled NASA to test faster and smarter, according to the agency. Each part was built by layering metal powder which was fused together with a laser. Whereas with a traditional manufacturing method 163 individual parts would be made and assembled, with 3D printing only two parts were required!

completely printed, the bed is lowered by one layer thickness and new layer of fused material is applied. The process is repeated until all the fused layers have been laid. Fig. 3 shows an SLS system.

Stereolithography (SLA). This method employs ultraviolet curable photopolymer resin and an ultraviolet laser to build the object's layers. The laser beam scans a selective surface area of the resin to solidify it as per the 3D design model. Once a layer is printed, the platform descends by a distance equal to the thickness of a single layer. And the laser scans again to print the second layer. Fig. 4 shows and SLA system.

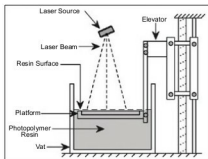


Fig. 4: SLA system

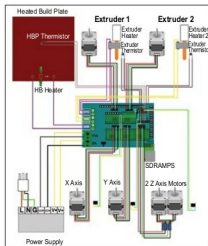


Fig. 5: Electronic and electrical system



Fig. 6: Stepper motors



Fig. 7: Endstop

How to build a DIY 3D printer

The printer described here is the LM8UU Prusa Mendel 3D printer. The basics of building any FDM based DIY printer will be more or less the same. The printer has three major parts: electronics (and electrical), software and mechanics.

Fig. 5 shows the complete electronic and electrical system that runs a 3D printer. Sanguinololu board is the heart of the whole system. It controls different stepper motors for moving the nozzle in X and Y directions, moving the heat bed in Z direction and extruding the material from the nozzle.

Fig. 6 shows the stepper motors used. The board senses extruder's and bed's temperature through thermis-

tors. Endstops are used (like stoppers) by the printer to determine the boundaries. Fig. 7 shows an endstop.

The temperatures of nozzle and heat-bed can be monitored through



Fig. 8: ATX power supply

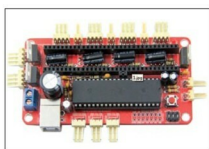


Fig. 9: Sanguinololu board



Fig. 10: Pololu board with heatsink



Fig. 11: Heat-bed

software as explained in software section. The software program, which connects to the board through USB interface, is used for loading the 3D model file for printing and testing